determine them with the current approximate equations, physicists have naturally studied the physics that follows from a range of possibilities. This is represented in Figure 12.1 by the shaded regions —each point in such a region denotes one specific choice for the coupling constant and the curled-up geometry. Without invoking any dualities, we still have five disjointed (collections of) theories.

But now, if we apply all of the dualities we have discussed, then as we vary the coupling and geometric parameters, we can pass from any one theory to any other, so long as we also include the unifying central region of M-theory; this is shown in Figure 12.2. Even though we have only a scant understanding of M-theory, these indirect arguments lend strong support to the claim that it provides a unifying substrate for our five naively distinct string theories. Moreover, we have learned that M-theory is closely related to yet a sixth theory—eleven-dimensional supergravity—and this is recorded in Figure 12.11, a more precise version of Figure 12.2.13

Figure 12.11 illustrates that the fundamental ideas and equations of M-theory, although only partially understood at the moment, unify those of all of the formulations of string theory. M-theory is the theoretical elephant that has opened the eyes of string theorists to a far grander unifying framework.

A Surprising Feature of M-Theory: Democracy in Extension

When the string coupling constant is small in any of the upper five peninsular regions of the theory map in Figure 12.11, the fundamental ingredient of the theory appears to be a one-dimensional string. We have, however, just gained a new perspective on this observation. If we start in either the Heterotic-E or Type IIA regions and turn the value of the respective string coupling constants up, we migrate toward the center of the map in Figure 12.11, and what appeared to be one-dimensional strings stretch into two-dimensional membranes. Moreover, through a more or less intricate sequence of duality relations involving both the string coupling constants and the detailed form of the curled-up spatial dimensions, we can smoothly and continuously move from any point in Figure 12.11 to any other. Since the two-dimensional membranes we have come upon from the Heterotic-E and Type IIA perspectives can be followed as we migrate to any of the three other string formulations in Figure 12.11, we learn